TQM UNIT III — QUALITY FUNCTION DEPLOYMENT

40-Minute Exam Preparation Notes | Based on the uploaded 30-slide QFD PPT

40-minute plan

Time	Study focus	PPT slides
0–5 min	Definition, introduction and need for QFD	2–4
5–10 min	Four phases and benefits	5–10
10–15 min	Voice of the Customer and information types	11–13
15–28 min	House of Quality and QFD matrix	14–26
28–33 min	Summary and example	27–28
33–40 min	Recall key points and practise questions	Revision

1. Quality Function Deployment (QFD)

Definition: QFD is a systematic and organised approach that considers customer needs and demands when designing new products and services or improving existing ones.

Introduced: In Japan in 1966 by Yoji Akao.

Why is QFD needed?

- Understand customer needs and expectations.
- Carry customer requirements throughout product development and manufacturing.
- Improve product design and manufacturing.
- Reduce design mistakes and unnecessary engineering changes.
- Improve customer satisfaction.

Example: Laptop customers may want long battery life, low weight and high performance. QFD translates these needs into measurable engineering requirements.

2. Four phases of QFD

Memorise the sequence: **Product → Parts → Process → Production.**

Phase	Name	Transformation
I	Product Planning	Customer needs → Design requirements
II	Part Deployment	Design requirements → Parts requirements
III	Process Planning	Parts requirements → Process requirements
IV	Production Planning	Process requirements → Production standards

The four phases form a chain: customer needs are converted into design requirements, then parts, processes and finally production standards.

3. Benefits of QFD

Benefit	Points to write
Customer-driven	Understands customer needs, uses competitive information, prioritises resources.

Benefit	Points to write
Reduces implementation time	Decreases midstream design changes, limits post-introduction problems, avoids duplicated development.
Promotes teamwork	Improves communication, builds consensus and coordinates departments.
Provides documentation	Records design rationale, structures information and supports future changes.

Other benefits from the PPT: shorter product-development cycles, reduced start-up costs, fewer engineering changes, fewer design oversights, consensus decisions and written documentation.

4. Voice of the Customer (VOC)

Definition: The Voice of the Customer represents customer needs, expectations and desired product attributes. It is the driving force behind QFD.

- Customer requirements determine product attributes.
- Customer satisfaction means meeting or exceeding expectations.
- Expectations may be vague or general.
- Interpret requirements as the customer intends them, not as the organisation prefers.

Example: “I want a lightweight laptop” can be translated into a measurable weight requirement.

Types of customer information

Type	Example in the PPT
Solicited, measurable, routine	Customer and market surveys; trade trials
Unsolicited, measurable, routine	Customer complaints; lawsuits
Solicited, subjective, routine	Focus groups
Solicited, subjective, haphazard	Trade/customer visits; independent consultants
Unsolicited, subjective, haphazard	Conventions; vendors; suppliers

Solicited = actively requested; unsolicited = received without being specifically requested. Measurable information is quantifiable; subjective information reflects opinions or perceptions.

5. House of Quality (HOQ)

Definition: The House of Quality is a QFD matrix that connects customer requirements (WHATs) with technical descriptors (HOWs) to help prioritise technical actions.

Main components

Component	Meaning
Customer requirements (WHATs)	What the customer wants.
Technical descriptors (HOWs)	How the organisation will satisfy customer needs.
Correlation matrix (roof)	Relationships among technical descriptors.
Relationship matrix	Strength of relationship between WHATs and HOWs.
Customer competitive assessment	Comparison of customer-perceived performance with competitors.
Technical competitive assessment	Comparison of technical performance with competitors.
Prioritised customer requirements	Ranked customer needs.

Component	Meaning
Prioritised technical descriptors	Ranked technical actions or characteristics.

Relationship matrix scores (WHATs vs HOWs)

Score	Relationship
+9	Strong
+3	Medium
+1	Weak

Correlation matrix scores (HOWs vs HOWs)

Score	Relationship
+9	Strong positive
+3	Positive
–3	Negative
–9	Strong negative

How to draw the House of Quality in an exam

- Draw a house-shaped roof above a rectangular matrix.
- Place technical descriptors (HOWs) along the top.
- Place customer requirements (WHATs) down the left.
- Label the main rectangle as the relationship matrix.
- Label the roof as the correlation matrix.
- Add competitive assessment and prioritisation sections at the bottom and right.

6. Seven steps for building the House of Quality

1. List customer requirements (WHATs).

2. List technical descriptors (HOWs).

3. Develop relationships between WHATs and HOWs.

4. Develop interrelationships among HOWs.

5. Conduct competitive assessments.

6. Prioritise customer requirements.

7. Prioritise technical descriptors.

Memory sequence: Customer needs → Technical solutions → Relationships → Correlations → Competition → Customer priority → Technical priority.

7. Prioritisation and competitive assessment

- **Customer competitive assessment:** Evaluates how well the product meets customer needs compared with competitors.
- **Technical competitive assessment:** Compares technical performance against competing products.
- **Importance rating:** Indicates how important each customer requirement is.
- **Target value:** Desired performance level.

- **Scale-up factor:** Reflects the improvement required to reach the target.
- **Sales point:** Represents the sales importance assigned to a requirement.
- **Absolute weight:** Helps rank customer requirements by importance.

For conceptual questions, explain the purpose of each factor. For numerical questions, follow the values and formula specified in the question or class example.

8. Basic QFD weighted-priority calculation

A commonly used weighted-priority calculation is:

$$W_j = \Sigma (I_i \times R_{ij})$$

Where W_j is the priority weight of technical descriptor j; I_i is the importance rating of customer requirement i; and R_{ij} is the relationship score, typically 9, 3, 1 or 0 if no relationship is assigned.

Example: Importance rating = 5 and relationship = strong (9). Weight contribution = 5 × 9 = **45**. For multiple requirements, calculate each product and add them to obtain the technical descriptor's total weight.

Note: This is the standard weighted-priority method. Include scale-up or sales-point factors only when the problem explicitly requires them.

9. QFD summary

- Provides an orderly way to collect and present information.
- Shortens the product-development cycle.
- Can reduce start-up costs and engineering changes.
- Reduces the chance of design oversights.
- Encourages teamwork and consensus decisions.
- Preserves design decisions in writing.

Slide 28 contains a QFD example showing how customer requirements, technical descriptors and relationship scores come together in a completed matrix.

10. Expected exam questions

Possible 10-mark questions

- Explain QFD, its need, benefits and four phases.
- Draw and explain the House of Quality with its major components.
- Explain the steps involved in building a House of Quality.
- Explain the Voice of the Customer and classify the types of customer information.

Possible 5-mark questions

- Define QFD and explain its benefits.
- List and explain the four phases of QFD.
- What is the Voice of the Customer?
- Differentiate customer and technical competitive assessment.
- Explain the relationship matrix and correlation matrix with their scoring values.
- List the seven steps for building the House of Quality.

These are study predictions based on the PPT, not confirmed exam questions.

Final 5-minute recall checklist

- QFD definition; Yoji Akao; Japan, 1966.

• Four phases: Product → Parts → Process → Production.

• Four benefit categories: Customer-driven, faster implementation, teamwork, documentation.

• VOC and five types of customer information.

• House of Quality diagram and its components.

• Relationship scores: 9, 3, 1; correlation scores: +9, +3, –3, –9.